Event-Triggered Tube-DMPC With Shrinking Ingredients for Vehicle Platoon Under Disturbance and Communication Delay

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***Abstract*— This paper presents a novel event-triggered robust distributed model predictive control (DMPC) methodology with a shrinking prediction horizon and shrinking terminal region for vehicle platoon control in the presence of external disturbances and communication delays. Using the framework of tube-based MPC, an information transmission triggering mechanism is designed which is determined by control plan changes for the nominal leading vehicle, to reduce the communication frequency. Then the follower vehicles only update their control inputs, prediction horizons and corresponding terminal sets when they receive new information from the leader, so as to reduce the computational burden. The proposed approach effectively compensates the information errors caused by the communication delays by utilizing the timestamp information. When the leader's control plan changes during the delay period, the recursive feasibility of the local optimization problem and the uniform ultimate bound stability of the platoon system are guaranteed with the proposed controller. The numerical simulation demonstrates the superior performance and robustness of the proposed approach, which offers promising solutions for real-world platoon control systems.**

***Index Terms*— Communication delay, distributed model predictive control, event-triggered control, vehicle platoon.**

I. Introduction

Vehicle platooning, a group of vehicles following a leading vehicle in close spacing, has attracted great attention in recent decades thanks to its potential for improving traffic efficiency, energy consumption, and overall road safety [1], [2], [3]. As a typical kind of intelligent transportation system, vehicle platoons face a number of challenges before their practical application, including external disturbances, communication constraints, and computational burdens [4], [5]. Therefore, a suitable platoon control method should be designed to address these challenges efficiently and robustly.

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Model predictive control (MPC) has emerged as an effective control strategy for constrained dynamic systems [6], [7], [8]. It offers the capacity to take into account system constraints and optimize control inputs over a limited time horizon. By utilizing a system model to predict its future behaviour and compute optimal control inputs, MPC provides a predictive framework that aligns well with the requirements of platoon control. As a result, MPC has made great progress in the field of platoon control [9], [10], [11]. Distributed MPC (DMPC) strategies have significant advantages over centralized ones, which may suffer from heavy computational burdens when applied to multi-agent systems. In the framework of typical DMPC, the local optimization problem for each following vehicle is designed to achieve the overall control performance of the platoon [12].

In the platoon driving environment, heterogeneous external disturbances usually exist for each vehicle, such as the air resistance that is affected by neighbouring vehicles [13]. In [14], the tube-based MPC (Tube-MPC) framework in [15] is adopted to deal with lumped disturbances and to divide the vehicle system into a nominal and a real model for distributed controller design. This framework significantly reduces the burden of online computation when the control objective of the nominal optimization problem is constant and is easily combined with event-triggered controllers. An appropriate event-triggered transmission mechanism can be designed to reduce the communication and computation burden for multiagent systems [16], [17]. For example, state measurement errors are used as triggering conditions to reduce communication load [18], while other approaches trigger updates only when the state error exceeds a threshold [19]. However, these common triggering mechanisms are not directly compatible with tube-based DMPC methods, which focus on solving optimization problems for nominal systems.

Recent studies have further explored event-triggered mechanisms in distributed control systems under challenging conditions. For instance, [20] and [21] proposed distributed event-triggered MPC frameworks for vehicle platoons under DoS attacks, where dynamic thresholds and packet replenishment strategies are used to reduce communication frequency and ensure system stability. While these works primarily address security concerns, this paper focuses on the impact of control plan changes during communication delays, using a novel threshold based on the degree of control plan changes to reduce communication and computational burdens.

To further reduce the computational burden of MPC, one can consider changing the solution complexity of the optimization problem by varying the length of the prediction horizon [22]. A robust MPC with an adaptive shrinking prediction horizon as the state is gradually approaching the terminal region, is proposed in [23]. When these research are not aimed at distributed systems, an event-triggered DMPC with variable prediction horizon strategy is designed in [24] for spatially interconnected systems.

In addition to external disturbances, communication delays pose a significant challenge for multi-agent systems [25]. Traditional approaches often treat delays as unknown but bounded variables, focusing primarily on stability analysis without incorporating compensatory measures in controller design [26], [27]. This can lead to conservative designs that may not align with practical applications. A more practical approach is to transmit timestamps along with system state information during vehicle-to-vehicle communication, enabling compensation for discrepancies between current and delayed information [28], [29].

Recent studies have proposed various methods to address communication delays in platoon control. For instance, distributed sliding mode control has been used to handle homogeneous constant delays [30], while robust control strategies have been developed for heterogeneous time-varying delays [31]. Additionally, Lyapunov stability theory has been employed to determine the upper bound of communication delays, ensuring platoon stability [32]. These studies integrate delay parameters into the control law, ensuring stability and steering errors towards a bounded region near zero. However, most existing works assume a constant control plan for the leader [33], neglecting the impact of control plan changes during communication delays. This research fills this gap by providing a comprehensive analysis of the effects of such changes on platoon stability and performance.

To address these challenges and limitations, this paper proposes an innovative event-triggered robust DMPC approach. The key contributions are as follows:

1) Leveraging the MPC prediction framework and timestamp information, the proposed method compensates for discrepancies between delayed and current state information under the original control plan. It further addresses control plan changes during communication delays by calculating the invariant set of information errors and analyzing its relationship with recursive feasibility, stability, and ultimate error bounds.
2) Based on the Tube-DMPC framework, an event-triggered communication mechanism is designed, using the degree of changes in the nominal control plan as a threshold. This mechanism significantly reduces the frequency of data transmission and optimization problem solving, thereby alleviating both communication and computational burdens.
3) A prediction horizon update mechanism is proposed that dynamically adjusts based on the timing of received communication information. This mechanism not only further reduces the computational burden but also updates the corresponding shrinking terminal region.

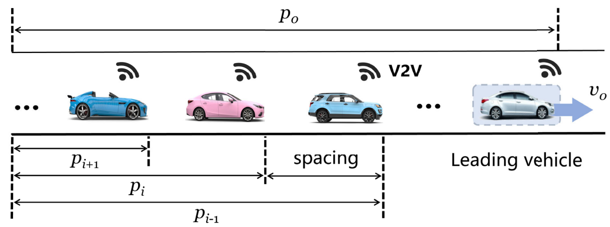

Fig. 1. A scenario and structure of the vehicle platoon.

4) The upper bound of communication delays is calculated under the conditions of recursive feasibility and Uniform Ultimate Boundedness (UUB) stability, given a specified desired error bounding range. This ensures that the system remains stable and within acceptable performance limits even in the presence of communication delays.

The remainder of this paper is organized as follows: Section II provides the problem formulation. An event-triggered DMPC with a shrinking prediction horizon is proposed in Section III. Section VI presents a detailed feasibility and stability analysis with the developed controller. The efficiency of the proposed method is verified by a numerical simulation in Section V and conclude the paper in Section VI.

Notation: Let $\mathbb{R}$ and $\mathbb{N}$ represent the set of reals and the set of nonnegative integers, respectively. Then, for integer $m,\ n \in \mathbb{N}$, $\mathbb{R}^{m\times n}$ is $m \times n$-dimensional real space. The interval (m, n) and $[m, n]$ denote the set $\{\xi \in \mathbb{R} \mid m < \xi < n\}$ and $\{\xi \in \mathbb{R} \mid m \le \xi \le n\}$, respectively. The box $([m_1, n_1], \ldots, [m_i, n_i])^\top$ is an interval vector. For $n_1, n_2 \in \mathbb{N}$ and $n_1 < n_2$, $\mathbb{N}_{\ge n_1} = \{n_1, n_1+1, \ldots\}$ and $\mathbb{N}_{[n_1,n_2]} = \{n_1, n_1+1, \ldots, n_2\}$. Given a symmetric matrix P, $P \succ 0$ means that the matrix P is positive definite. For a column vector x, $\|x\|$ and $\|x\|_P = \sqrt{x^T P x}$ denote the Euclidean norm and the P-norm, respectively. For matrix $P \in \mathbb{R}^{n\times n}$, constant $c \in \mathbb{R}$ and vector $\xi \in \mathbb{R}^n$, the set $\bar{\Omega}_P(c) = \{\xi \mid \|\xi\|_P < c\}$. The Minkowski sum and Pontryagin difference of two sets $\mathcal{X}$ and $\mathcal{Y}$ are given by $\mathcal{X} \oplus \mathcal{Y} = \{x + y \mid x \in \mathcal{X},\ y \in \mathcal{Y}\}$ and $\mathcal{X} \ominus \mathcal{Y} = \{x \mid x \oplus \mathcal{Y} \subseteq \mathcal{X}\}$, respectively.

II. Problem Formulation

A. Platoon Modeling

This paper considers a vehicle platoon consisting of a leader vehicle and some follower vehicles, which is shown in Figure 1. The longitudinal dynamic model of each vehicle can be formulated as follows:

$$\begin{aligned} p_i(k+1) &= p_i(k) + v_i(k)\Delta t \\ v_i(k+1) &= v_i(k) + \left(u_i(k) - \frac{f_i(v_i(k), \theta_i(k))}{m_i}\right)\Delta t \end{aligned} \tag{1}$$

where $p_i(k) \in \mathbb{R},\ v_i(k) \in \mathbb{R}$ represent the position and velocity of ith vehicle at discrete time $k \in \mathbb{N}_{\ge 0}$, the control input $u_i(k) \in \mathbb{R}$ represents the desired acceleration of the vehicle, $\Delta t \in \mathbb{R}$ is the discrete time interval. The comprehensive resistive force f_i is a function of the vehicle's velocity v_i and the road slope angle θ_i, incorporating aerodynamic drag, rolling resistance, and slope resistance [34], [35]. It is defined as

$$f_i(v_i, \theta_i) = C_{A,i} v_i^2 + c_r m_i g \cos(\theta_i) + mg \sin(\theta_i)$$

where c_r is the rolling resistance coefficient, m_i is the vehicle mass, g is the gravitational constant, and $C_{A,i} \in \mathbb{R}$ is the comprehensive air drag coefficient of the vehicle in platooning, whose value is affected by inter-vehicle spacing and cross-sectional areas of the adjacent vehicle [13].

By using $x_i = [p_i \ v_i]^\top$ to denote the system state of the ith vehicle, the system model can be rewritten in a state-space form as

$$x_i(k+1) = Ax_i(k) + Bu_i(k) + Hw_i(k) \tag{2}$$

where

$$A = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \Delta t \end{bmatrix}, \quad H = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

the term w_i, defined as the sum of the resistance acceleration $\frac{f_i}{m_i}$ and a stochastic disturbance or uncertainty term $\hat{w}_i$, is referred to as the lumped disturbance. It is assumed that (A, B) is stabilisable and the disturbance is amplitude-bounded, *i.e.*, $w_i(k) \in \mathcal{W}_i$, where $\mathcal{W}_i$ is a compact interval containing the origin in its interior.

The system is subject to state and control constraints

$$u_i \in \mathcal{U}_i, \ \ x_i \in \mathcal{X}_i, \tag{3}$$

where $\mathcal{U}_i \subset \mathbb{R}$ is compact, $\mathcal{X}_i \subset \mathbb{R}^2$ is closed, and each set contains the origin in its interior.

The vehicle system, without considering the bounded disturbance $w_i(k)$, can be treated as a nominal system:

$$\bar{x}_i(k+1) = A\bar{x}_i(k) + B\bar{u}_i(k). \tag{4}$$

Note that for any variable $s(k)$ and set $\mathcal{S}$ of actual vehicle system (2), $\bar{s}(k)$ and $\bar{\mathcal{S}}$, respectively, represent the corresponding variable and set of its nominal system (4).

B. Control Objective

The objective of platoon control is to track the leader's speed while preserving a desired gap between consecutive vehicles as stipulated by a designated spacing policy. This work specifically addresses scenarios where the platoon leader adjusts the control plan within the queue under external disturbances and communication delays. The term "control plan" refers to the system's planned sequence of control inputs for a future time interval, with the leader's control plan being transmitted to followers through a communication network. The leader's information can be transmitted to followers either directly or indirectly, based on the following assumption regarding the communication topology.

Assumption 1: The communication topology among the vehicles in the platoon contains a directed spanning tree.

Efficient inter-vehicle communication plays a significant role in platoon control. Nevertheless, it is essential to acknowledge that there exists a bounded communication delay $\tau_d \in \mathbb{N}_{[0,\bar{\tau}_d]}$ in the information transmission between each agent in the platoon, where $\bar{\tau}_d \in \mathbb{N}_{\geq 1}$ represents the maximum delay.

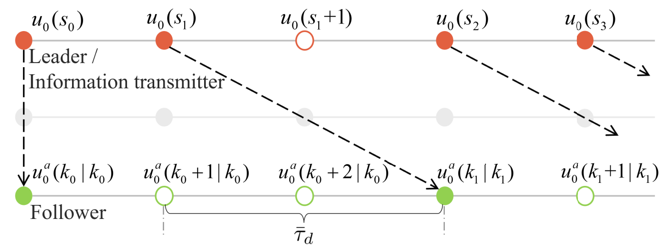

Fig. 2. Information transfer process with communication delay in the platoon.

As the formation control is to maintain the relative positions between each agent and the leader, the communication process in this work is that each agent acquires information from the leader, whether this information comes directly from the leader or is relayed through other agents based on various communication topologies. For simplicity, the worst-case scenario is considered in the following assumption.

Assumption 2: The maximum communication delay $\bar{\tau}_d \in \mathbb{N}_{\geq 1}$ is adopted for the agent who transfers the information.

Remark 1: The time at which the leader transmits data is denoted as $s_t \in \mathbb{N}_{\geq 0}$, and the moment at which the ith follower receives the leader's data is denoted as $k_{t,i} \in \mathbb{N}_{\geq 0}$, where $t \in \mathbb{N}_{\geq 0}$ is the order statistic. Therefore, the information reception time of the ith agent is represented as $k_{t,i} = k_t = s_t + \bar{\tau}_d$, $\forall t \in \mathbb{N}_{\geq 1}$. In addition, the initial control at time $s_0 = k_0 = 0$ happened after all the followers received the first data $\bar{x}_0^a(k_0) = \bar{x}_0(k_0) = x_0(k_0)$ from the leader vehicle, where $\bar{x}_0^a$ is the state variable of the leader assumed by the followers based on the information they have acquired.

Remark 2: The transmission process of information between the leader/information transmitter and follower with the maximum delay time $\bar{\tau}_d$, is shown in Figure 2. The $u_0^a(k_0+1 \mid k_0)$ represents the assumed input of the leader at time k_0+1 calculated based on the follower received information at time k_0. To accommodate the information transfer delay, advanced sensors installed in each vehicle encode transmitted information with precise timestamps. Subsequently, upon receiving the information, the sensors decode it, allowing the estimation of specific delay times by comparing it with the current time. The incorporation of timestamps allows for compensation of information errors arising from communication delays. However, it is essential to note that information errors resulting from changes in the leader's control plan cannot be anticipated or eliminated, as they are subject to the dynamic nature of the platooning environment.

In line with the discussions in Remarks 1 and 2, the compensation mechanism designed in this paper for the leader state information received by the follower, based on timestamp information and predictive models, is as follows:

$$\begin{aligned} \bar{x}_0^a(k+1 \mid k_t) &= A\bar{x}_0^a(k \mid k_t) + B\bar{u}_0^a(k \mid k_t) \\ \bar{u}_0^a(k \mid k_t) &= \bar{u}_0(k \mid s_t), \quad \forall k \in [k_t, k_{t+1}) \\ \bar{x}_0^a(k_t \mid k_t) &= A_p\bar{x}_0(s_t) + B_p\bar{U}_0^p(s_t), \end{aligned} \tag{5}$$

where $A_p = A^{\bar{\tau}_d}$, $B_p = [A^{\bar{\tau}_d - 1}B, \ldots, B]$ and $\bar{U}_0^p(s_t) = [\bar{u}_0(s_t \mid s_t), \ldots, \bar{u}_0(k_t - 1 \mid s_t)]^\top$.

With the above definition, one can define the platoon error between the leader vehicle and ith follower vehicle at time $k \in [k_t, k_{t+1})$, $\forall t \in \mathbb{N}_{\geq 0}$ as follows:

actual error: $e_i(k) = x_i(k) - x_0(k) + \hat{d}_i$

nominal error: $\bar{e}_i(k) = \bar{x}_i(k) - \bar{x}_0(k) + \hat{d}_i$

predicted error: $\bar{e}_i^p(k \mid k_t) = \bar{x}_i(k \mid k_t) - \bar{x}_0^a(k \mid k_t) + \hat{d}_i$

where $\hat{d}_i = [d_i \quad 0]^\top$, $d_i = i \cdot d_0$, and $d_0 \in \mathbb{R}$ is the desired space between adjacent vehicles, with the initial position $p_i > p_{i+1}$, $\forall i \in \mathbb{N}_{\geq 0}$. $\bar{e}_i^p(k \mid k_t)$ is the predicted error at time k calculated by the ith follower based on the information it received at time k_t. Similarly, define the predicted input error as

$$\bar{e}_{u,i}^p(k \mid k_t) = \bar{u}_i(k \mid k_t) - \bar{u}_0^a(k \mid k_t).$$

Then, the control objectives can be defined as

$$e_i(k) \in \sigma_i, \quad i \in \mathbb{N}_{\geq 1}, \; k \in \mathbb{N}_{\geq \hat{k}} \tag{6}$$

where $\sigma_i \subset \mathbb{R}^2$ is an invariant set and its size depends on the system disturbance and controller parameters design, and $\hat{k} \in \mathbb{N}_{\geq 1}$ is a positive integer, both values will be made clear in Section IV.

III. Event-Triggered Tube-DMPC With Shrinking Prediction Horizon

In this section, an event-triggered Tube-DMPC with shrinking prediction horizon is designed for a vehicle platoon with the nominal predicted system to achieve the stability of the platoon and the control objectives.

A. Distributed Controller Design

Consider the control performance of platoon error and acceleration penalty, the cost function of the distributed MPC algorithm can be defined as

$$\begin{aligned} &\mathscr{J}_i(\bar{x}_i(k_t), \bar{x}_0^a(k_t), \bar{U}_i(k_t), N_i(k_t)) \\ &= \sum_{\tau=0}^{N_i(k_t)-1} J_i(\bar{x}_i(k_t+\tau \mid k_t), \bar{u}_i(k_t+\tau \mid k_t)) \\ &\quad + \|\bar{e}_i^p(k_t+N_i(k_t) \mid k_t)\|_{P_i}^2, \end{aligned} \tag{7}$$

where $N_i(k_t) \in \mathbb{N}_{\geq 1}$ is the prediction horizon at time k_t,

$$\begin{aligned} J_i(\bar{x}_i(k_t+\tau \mid k_t), \bar{u}_i(k_t+\tau \mid k_t)) = \|\bar{e}_i^p(k_t+\tau \mid k_t)\|_{Q_i}^2 \\ + \|\bar{e}_{u,i}^p(k_t+\tau \mid k_t)\|_{R_i}^2, \end{aligned}$$

$\bar{U}_i(k_t) = \{\bar{u}_i(k_t \mid k_t), \bar{u}_i(k_t+1 \mid k_t), \ldots, \bar{u}_i(k_t+N_i(k_t)-1 \mid k_t)\}$ and $Q_i > 0$, $R_i > 0$, $P_i > 0$ are the weighting matrices.

The safety constraints for vehicle platoons should also be guaranteed during the control process with the following constraints:

$$\begin{aligned} e_{p,i}(k_t+\tau \mid k_t) &= [1 \; 0]e_i(k_t+\tau \mid k_t) \\ &= p_i(k_t+\tau \mid k_t) - p_0(k_t+\tau \mid k_t) + d_i \\ &\in [-p_b, \; p_b] \\ e_{v,i}(k_t+\tau \mid k_t) &= [0 \; 1]e_i(k_t+\tau \mid k_t) \\ &= v_i(k_t+\tau \mid k_t) - v_0(k_t+\tau \mid k_t) \\ &\in [-v_b, \; v_b] \end{aligned} \tag{8}$$

for $\tau \in \mathbb{N}_{[0, N_i(k_t)]}$, where $p_b \in (0, \infty)$ and $v_b \in (0, \infty)$ are the boundary parameters of the position error and velocity error, respectively. The constraints in (8) can be rewritten in a compact form

$$e_i(k_t+\tau \mid k_t) \in \hat{\mathcal{E}}_i, \tag{9}$$

where the box $\hat{\mathcal{E}}_i = ([-p_b, p_b], [-v_b, v_b])^\top$.

To constrain the actual system state with a Tube-MPC framework, one needs to identify the relation between the nominal constraints and the actual ones. First define the control law of actual system (2) as

$$u_i(k_t) = \bar{u}_i(k_t) + K\zeta_i(k_t) \tag{10}$$

where K is a feedback gain that satisfies $A_K = A + BK$ is Schur stable, $\zeta_i(k_t) = x_i(k_t) - \bar{x}_i(k_t)$.

Then, a minimal positive invariant set $\mathcal{Z}_{w_i}$ can be found for ζ_i with bounded disturbance w_i. That is, $\forall \zeta_i(k_t) \in \mathcal{Z}_{w_i}$, the successor state $\zeta_i(k_t+1) = A_K\zeta_i(k_t) + w_i \in \mathcal{Z}_{w_i}$, where

$$\mathcal{Z}_{w_i} = \bigoplus_{j=0}^{\infty} A_K^j \mathcal{W}_i, \tag{11}$$

The infinite Minkowski sum can be outer approximated by using the method introduced in [14].

Assumption 3: The difference between the control inputs of the leader vehicle at adjacent discrete time instant s_t and $s_t - 1$ satisfies $\|\bar{u}_0(s_t+\tau \mid s_t) - \bar{u}_0(s_t+\tau \mid s_t-1)\| \leq \delta_0(\tau)$, for $\tau \in \mathbb{N}_{[0, N_i(k_t)]}$, where $\delta_0(\tau) \in (0, \infty)$ is the bound of the difference. In addition, $\bar{u}_0 \in \bar{\mathcal{U}}_0$, and $\bar{\mathcal{U}}_0 \oplus K\bar{e}_i^p \in \bar{\mathcal{U}}_i$, $\forall \bar{e}_i^p \in \bar{\Omega}_{P_i}(\hat{\alpha}_i)$, where $\hat{\alpha}_i \in (0, \infty)$.

Then, the change range of the leader's control law at step $r \in \mathbb{N}_{\geq 0}$ of prediction can be represented as

$$\mathcal{Z}_{\delta_0}(r) = \{\eta \in \mathbb{R} \mid \|\eta\| \leq \delta_0(r)\}.$$

Therefore, when the leader changes its control plan within the set $\mathcal{Z}_{\delta_0}$, the state error at step r of prediction between the original plan and actual state will be confined within a set as

$$\mathcal{Z}_{\hat{\delta}_0}(r) = \bigoplus_{r_1=0}^{r} A^{r_1} B \mathcal{Z}_{\delta_0}(r - r_1).$$

Correspondingly, to describe the degree of change in the leader's state and to be used in the design of triggering mechanisms in subsequent work, define the following set:

$$\mathcal{Z}_\beta(\tau) = \beta \mathcal{Z}_{\hat{\delta}_0}(\tau) \tag{12}$$

where $\beta \in (0, 1)$ is a scaling factor. The set $\mathcal{Z}_\beta(\tau)$ is used to adjust the sensitivity of the communication to changes in the leader's state. This definition allows us to design a flexible and adaptive triggering mechanism that balances communication efficiency and control performance.

Considering the worst-case scenario, if the leader continuously changes its control plan until the follower receives the information with $\bar{\tau}_d$ steps delay, then the error between the state of the leader after $\tau \in \mathbb{N}_{[0, N_i(k_t)]}$ steps in the future as estimated by the follower and the actual state of the leader will reside within the following set:

$$\mathcal{Z}_{\Delta_0}(\tau, \bar{\tau}_d) = \bigoplus_{r=\tau}^{\tau+\bar{\tau}_d-1} \mathcal{Z}_{\hat{\delta}_0}(r). \tag{13}$$

Then, based on (2), (3), (10), (11), (12) and (13), the constraints of nominal system (4) can be computed as

$$\begin{aligned} \bar{x}_i &\in \bar{\mathcal{X}}_i = \mathcal{X}_i \ominus \mathcal{Z}_{w_i} \\ \bar{u}_i &\in \bar{\mathcal{U}}_i = \mathcal{U}_i \ominus K_i \mathcal{Z}_{w_i} \\ \bar{e}_i^p(k_t+\tau \mid k_t) &\in \bar{\mathcal{E}}_i^p(\tau) = \mathcal{E}_i(\tau) \ominus \mathcal{Z}_{\mathcal{E}_i}(\tau, \bar{\tau}_d) \\ \bar{e}_i^p(k_0+N_i(k_0) \mid k_0) &\in \bar{\Omega}_{P_i}(\gamma_i \hat{\alpha}_i) \\ &\subseteq \bar{\Omega}_{P_i}(\gamma_i \hat{\rho}_i) \ominus \mathcal{Z}_{\mathcal{E}_i}(N_i(k_0), \bar{\tau}_d) \end{aligned} \tag{14}$$

where $\mathcal{E}_i(\tau)$ is a subset of $\hat{\mathcal{E}}_i$, $\hat{\rho}_i \in (\hat{\alpha}_i, \infty)$ is used to described the size of actual system's terminal set, and $\mathcal{Z}_{\mathcal{E}_i}(\tau, \bar{\tau}_d) = \mathcal{Z}_{w_0} \oplus \mathcal{Z}_{w_i} \oplus \mathcal{Z}_{\Delta_0}(\tau, \bar{\tau}_d) \oplus \mathcal{Z}_\beta(\tau+\bar{\tau}_d)$, $\forall \tau \in \mathbb{N}_{[0, N_i(k_t)]}$.

Considering the inaccessibility of actual states and real-time information during the prediction and control processes, a local optimization problem is designed with the predicted error $\bar{e}_i^p(k)$ and the above nominal constraints as follows:

$$\textbf{OP}: \quad \min_{\bar{U}_i(k_t)} \mathscr{J}_i(\bar{x}_i(k_t), \bar{x}_0^a(k_t), \bar{U}_i(k_t), N_i(k_t)) \tag{15a}$$

$$\begin{aligned} \text{s.t. } \bar{e}_i^p(k_t+\tau+1 \mid k_t) &= A\bar{e}_i^p(k_t+\tau \mid k_t) \\ &\quad + B\bar{e}_{u,i}^p(k_t+\tau \mid k_t) \end{aligned} \tag{15b}$$

$$\bar{e}_i^p(k_t+\tau \mid k_t) \in \bar{\mathcal{E}}_i^p(\tau), \ \ \tau \in \mathbb{N}_{[0,N_i(k_t)]} \tag{15c}$$

$$\bar{e}_i^p\left(k_t+N\left(k_t\right) \mid k_t\right) \in \bar{\Omega}_{P_i}\left(\gamma_i \hat{\alpha}_i\right) \tag{15d}$$

$$\bar{x}_i\left(k_t+\tau \mid k_t\right) \in \bar{\mathcal{X}}_i, \qquad \tau \in \mathbb{N}_{[0,N_i(k_t)]} \tag{15e}$$

$$\bar{u}_i\left(k_t+\tau \mid k_t\right) \in \bar{\mathcal{U}}_i, \qquad \tau \in \mathbb{N}_{[0,N_i(k_t)-1]} \tag{15f}$$

where $\gamma_i \in (0,1)$, $\hat{\alpha}_i$ is the parameter of terminal set which will be designed and discussed in Section IV. $\bar{U}_i$ is the sequence of nominal control inputs $\bar{u}_i$, $\bar{x}_i$ is the nominal state computed based on the nominal dynamics (4), and the nominal optimal control input and system state sequences for ith vehicle at time k_t are represented as $\bar{U}_i^*(k_t) = \{\bar{u}_i^*(k_t \mid k_t), \bar{u}_i^*(k_t+1 \mid k_t), \ldots, \bar{u}_i^*(k_t+N_i(k_t)-1 \mid k_t)\}$ and $\bar{X}_i^*(k_t) = \{\bar{x}_i^*(k_t \mid k_t), \bar{x}_i^*(k_t+1 \mid k_t), \ldots, \bar{x}_i^*(k_t+N_i(k_t) \mid k_t)\}$, respectively. Additionally, the error $\bar{e}_i^p$ is defined as the error between the nominal state $\bar{x}_i$ and the compensated state $\bar{x}_0^a$ of the leader with a desired spacing d_i, ensuring that the system tracks the desired trajectory. The terminal region $\bar{\Omega}_{P_i}(\gamma_i\hat{\alpha}_i)$ satisfies $\bar{\mathcal{X}}_0 \oplus \bar{\Omega}_{P_i}(\gamma_i\hat{\alpha}_i) \subseteq \bar{\mathcal{X}}_i$ and $\bar{\Omega}_{P_i}\left(\gamma_i\hat{\alpha}_i\right) \subseteq \bar{\mathcal{E}}_i^p(N_i(k_0))$.

For simplicity and without loss of generality, the following assumption is made.

Assumption 4: Problem **OP** for the ith vehicle at initial time is feasible.

B. Triggering Mechanism and Prediction Horizon Updating Strategy

Due to the terminal constraints, the proposed controller may need a large prediction horizon $N_i(k_0)$ at the initial step $k_0 = 0$ to handle the situation where the initial platoon error $\bar{e}_i^p(0)$ is far from the terminal region $\bar{\Omega}_{P_i}(\gamma_i\hat{\alpha}_i)$. Therefore, in this section, a shrinking prediction horizon strategy with a triggering mechanism for the controller is proposed to reduce its computation burden as the platoon error gradually enters the terminal invariant set.

With Remarks 1, 2 and equation (5), the assumed leader's information $\bar{x}_0^a(k)$ at time $k > k_t$ for followers can be obtained as follows: 1) if new information is received, $k_{t+1} = k$, and $\bar{x}_0^a(k+\tau \mid k) = \bar{x}_0(k+\tau \mid s_{t+1})$, $\tau \in \mathbb{N}_{[0,N_i(k_{t+1})]}$; 2) otherwise, $\bar{x}_0^a(k+\tau \mid k) = \bar{x}_0^a(k+\tau \mid k_t)$, $\tau \in \mathbb{N}_{[0,N_i(k_t)]}$, for $t \in \mathbb{N}_{\geq 0}$, and $\bar{x}_0^a(k_0) = \bar{x}_0(k_0)$. The follower's controller is triggered only when it receives new information from the leader.

The data transmission condition of the leader is influenced by difference between the current system state $\bar{x}_0(s_{t+1})$ and the last transmitted state $\bar{x}_0(s_{t+1} \mid s_t)$. That means the triggering condition will be satisfied at time $s_{t+1} > s_t$, $t \in \mathbb{N}_{\geq 0}$ when

$$\begin{aligned} &\bar{x}_0(s_{t+1}+\tau \mid s_{t+1}) - \bar{x}_0(s_{t+1}+\tau \mid s_t) \notin \mathcal{Z}_\beta(\tau), \\ &\tau \in \mathbb{N}_{[0,\, N_0(k_0)]} \end{aligned} \tag{16}$$

or

$$s_{t+1} = s_t + N_0\left(k_0\right) \tag{17}$$

where the size of $\mathcal{Z}_\beta(\tau)$ can be affected by choosing the parameter β.

Therefore, based on the control input constraint of the leader in Assumption 3, it has

$$\bar{x}_0(s_{t+1}+\tau \mid s_{t+1}) - \bar{x}_0(s_{t+1}+\tau \mid s_t) \in \mathcal{Z}_{\hat{\delta}_0,\beta}(\tau), \tag{18}$$

where $\mathcal{Z}_{\hat{\delta}_0,\beta}(\tau) = \mathcal{Z}_{\hat{\delta}_0}(\tau) \oplus \mathcal{Z}_\beta(\tau+1)$, for $\tau \in \mathbb{N}_{[0,N_i(k_{t+1})]}$.

Then, one can get the region of state difference between the follower received information at adjacent steps:

$$\bar{x}_0^a(k_{t+1}+\tau \mid k_{t+1}) - \bar{x}_0^a(k_{t+1}+\tau \mid k_t) \in \mathcal{Z}_{\hat{\delta}_0,\beta}(\tau+\bar{\tau}_d) \tag{19}$$

The new prediction horizon $N_i(k_{t+1})$ can be calculated at transmission triggering time s_{t+1} based on the last transmission triggering time s_t and last prediction horizon $N_i(k_t)$ as

$$N_i(k_{t+1}) = N_i(k_t) + s_t - s_{t+1} + 1, \ t \in \mathbb{N}. \tag{20}$$

In this work, dual-mode MPC strategy is adopted. Referring to the definition of system errors, design the terminal feedback control law as $\bar{u}_i^p(k) = \bar{u}_0^a(k) + K(\bar{x}_i(k) - \bar{x}_0^a(k))$ when $N_i(k) = 1$ is satisfied. With the designed control law, the dynamics of the predicted state error between the leader and the ith follower can be described as $\bar{e}_i^p(k+1) = A_K\bar{e}_i^p(k)$.

IV. Feasibility and Stability Analysis

In this section, the proof of recursive feasibility and stability with the proposed method for the nominal platoon system is provided to verify the effectiveness of Algorithm 1.

Firstly it needs to prove that problem **OP** has a feasible solution and satisfies the constraints at time k_{t+1} when the problem is feasible at time k_t, for $t \in \mathbb{N}_{\geq 0}$. Therefore, a feasible control sequence $\tilde{U}_i(k_{t+1}) = \{\tilde{u}_i(k_{t+1} \mid k_{t+1}), \tilde{u}_i(k_{t+1}+1 \mid k_{t+1}), \ldots, \tilde{u}_i(k_{t+1}+N_i(k_{t+1}) \mid k_{t+1})\}$ for the ith vehicle at time k_{t+1} can be defined, which can be obtained with the last optimal solution $\tilde{U}_i^*(k_t)$ by

$$\begin{aligned} &\tilde{u}_i\left(k_{t+1}+\tau \mid k_{t+1}\right) \\ &= \begin{cases} \bar{u}_i^*\left(k_{t+1}+\tau \mid k_t\right), \ \ \tau \in \mathbb{N}_{[0,N_i(k_{t+1})-2]} \\ \bar{u}_0^a\left(k_{t+1}+\tau \mid k_{t+1}\right) + K\left(\tilde{x}_i(k_{t+1}+\tau \mid k_{t+1})\right. \\ \left. -\bar{x}_0^a(k_{t+1}+\tau \mid k_{t+1})\right), \ \ \tau = N_i(k_{t+1})-1 \end{cases} \end{aligned} \tag{21}$$

Algorithm 1 Event-Triggered DMPC With Shrinking Ingredients. (⊚: Leader, ⊖: Follower)

Step 1. At initial time $k = s_0 = k_0 = t = 0$:
⊚: Send $\bar{X}_0(0)$, $\bar{U}_0(0)$ to followers, apply $\bar{u}_0(s_0 \mid s_0)$ to system until $k = 1$;
⊖: Receive $\bar{X}_0^a(0)$, $\bar{U}_0^a(0)$ from leader, solve problem **OP** to get $\bar{U}_i^*(0)$, finally implement $\bar{u}_i^*(k_0 \mid k_0)$ to system until $k = 1$.
Step 2. At time $k \geq 1$:
⊚: If triggering condition (16) or (17) is satisfied, go to Step 3, otherwise, go to Step 4;
⊖: If ith vehicle receive the information from leader, go to Step 3, otherwise, go to Step 4.
Step 3.
⊚: Set $s_{t+1} = k$, send $\bar{X}_0(s_{t+1})$, $\bar{U}_0(s_{t+1})$ to followers, apply $\bar{u}_0(s_{t+1} \mid s_{t+1})$ to system; set $t := t + 1$;
⊖: Set $k_{t+1} = k$, receive $\bar{X}_0^a(k_{t+1})$, $\bar{U}_0^a(k_{t+1})$ from leader, use (20) to update $N_i(k_{t+1})$, if $N_i(k_{t+1}) = 1$, apply $\bar{u}_i(k_{t+1} \mid k_{t+1}) = \bar{u}_0^a(k_{t+1} \mid k_{t+1}) + K(\bar{x}_i(k_{t+1} \mid k_{t+1}) - \bar{x}_0^a(k_{t+1} \mid k_{t+1}))$, otherwise, solve **OP** to get $\bar{U}_i^*(k_{t+1})$ and implement $\bar{u}_i^*(k_{t+1} \mid k_{t+1})$; set $t := t + 1$, go to Step 5.
Step 4.
⊚: Apply $\bar{u}_0(k \mid k)$ to system;
⊖: Set $N_i(k) = N_i(k-1)$, apply $\bar{u}_i(k \mid k) = \bar{u}_i^*(k \mid k_t)$.
Step 5.
Set $k := k + 1$, and return to Step 2.

The corresponding feasible state sequence is $\tilde{X}_i(k_{t+1}) = \{\tilde{x}_i(k_{t+1} \mid k_{t+1}), \tilde{x}_i(k_{t+1} + 1 \mid k_{t+1}), \ldots, \tilde{x}_i(k_t + N_i(k_{t+1}) \mid k_{t+1})\}$, which can be calculated with the feasible control input as

$$\tilde{x}_i\,(k_{t+1} + \tau + 1 \mid k_{t+1}) = A\tilde{x}_i(k_{t+1} + \tau \mid k_{t+1}) + B\tilde{u}_i(k_{t+1} + \tau \mid k_{t+1}) \tag{22}$$

where $\tau \in \mathbb{N}_{[0,N_i(k_{t+1})-1]}$, $\tilde{x}_i\,(k_{t+1} \mid k_{t+1}) = \bar{x}_i^*\,(k_{t+1} \mid k_t)$.

Based on the above definition, it is clear that the feasible solution and state satisfy the constraints (15e) and (15f) in the nominal **OP**, that is,

$$\begin{aligned} &\bar{x}_i\,(k_{t+1} + \tau \mid k_{t+1}) \in \bar{\mathcal{X}}_i, \ \ \tau \in \mathbb{N}_{[0,N_i(k_{t+1})-1]} \\ &\bar{u}_i\,(k_{t+1} + \tau \mid k_{t+1}) \in \bar{\mathcal{U}}_i, \ \ \tau \in \mathbb{N}_{[0,N_i(k_{t+1})-2]} \end{aligned}$$

while the constraints (15c) and (15d) will be guaranteed with the following theorem.

Theorem 1: For the ith follower, $i \in \mathbb{N}_{\geq 1}$, suppose Assumptions 1–4 hold, and $N\,(k_0) \geq 2$. The problem **OP** *has a feasible solution at time instant k_{t+1} for $t \in \mathbb{N}_{\geq 0}$, if the following conditions hold*:

i) $\bar{\mathcal{E}}_i^p(\tau + 1) \oplus \mathcal{Z}_{\hat{\delta}_0,\beta}(\tau + \bar{\tau}_d) \subseteq \bar{\mathcal{E}}_i^p(\tau)$, $\tau \in \mathbb{N}_{[0,N_i(k_0)-1]}$;
ii) $\mathcal{Z}_{\hat{\delta}_0,\beta}(N_i(k_0) + \bar{\tau}_d) \subseteq \bar{\Omega}_{P_i}((1 - \gamma_i)\,\hat{\alpha}_i)$;
iii) $\gamma_i^2 P_i - A_K^T P_i A_K \succ 0$.

Proof: For system i, it is assumed that problem **OP** has an optimal solution $\bar{U}_i^*(k_t)$ and corresponding sequence of states $\bar{X}_i^*(k_t)$. Then, with the defined control sequence $\tilde{U}_i(k_{t+1})$ and state sequence $\tilde{X}_i(k_{t+1})$, the corresponding error state is then

$$\begin{aligned} &\tilde{e}_i^p\,(k_{t+1} + \tau \mid k_{t+1}) \\ &= \tilde{x}_i\,(k_{t+1} + \tau \mid k_{t+1}) - \bar{x}_0^a\,(k_{t+1} + \tau \mid k_{t+1}) \\ &= \bar{x}_i^*\,(k_{t+1} + \tau \mid k_t) - \bar{x}_0^a\,(k_{t+1} + \tau \mid k_t) \\ &\quad + \bar{x}_0^a\,(k_{t+1} + \tau \mid k_t) - \bar{x}_0^a\,(k_{t+1} + \tau \mid k_{t+1}) \\ &= \bar{e}_i^p\,(k_{t+1} + \tau \mid k_t) \\ &\quad + (\bar{x}_0^a(k_{t+1} + \tau \mid k_t) - \bar{x}_0^a(k_{t+1} + \tau \mid k_{t+1})) \end{aligned} \tag{23}$$

Therefore, one can calculate the error between the error states $\tilde{e}_i^p\,(k_{t+1} + \tau \mid k_{t+1})$ and $\bar{e}_i^p\,(k_{t+1} + \tau \mid k_t)$ with equation (19) as

$$\begin{aligned} &\tilde{e}_i^p\,(k_{t+1} + \tau \mid k_{t+1}) - \bar{e}_i^p\,(k_{t+1} + \tau \mid k_t) \\ &= \bar{x}_0^a(k_{t+1} + \tau \mid k_t) - \bar{x}_0^a(k_{t+1} + \tau \mid k_{t+1}) \in \mathcal{Z}_{\hat{\delta}_0,\beta}(\tau + \bar{\tau}_d). \end{aligned} \tag{24}$$

According to equation (23) and condition i), it can be ensured that the error state $\tilde{e}_i^p\,(k_{t+1} + \tau \mid k_{t+1})$ at time $t + 1$ satisfies the constraint as

$$\begin{aligned} &\tilde{e}_i^p\,(k_{t+1} + \tau \mid k_{t+1}) \\ &= \bar{e}_i^p(k_{t+1} + \tau \mid k_t) \\ &\quad + (\bar{x}_0^a(k_{t+1} + \tau \mid k_t) - \bar{x}_0^a(k_{t+1} + \tau \mid k_{t+1})) \\ &\in \bar{\mathcal{E}}_i^p(\tau + k_{t+1} - k_t) \oplus \mathcal{Z}_{\hat{\delta}_0,\beta}(\tau + \bar{\tau}_d) \\ &\subseteq \bar{\mathcal{E}}_i^p(\tau), \ \ \tau \in \mathbb{N}_{[0,N_i(k_{t+1})-1]}. \end{aligned} \tag{25}$$

Furthermore, the constraint $\tilde{e}_i^p(k_{t+1} + N_i(k_{t+1}) \mid k_{t+1}) \subseteq \bar{\mathcal{E}}_i^p(N_i(k_{t+1}))$ can be guaranteed by proving that $\tilde{e}_i^p(k_{t+1} + N_i(k_{t+1}) \mid k_{t+1}) \in \bar{\Omega}_{P_i}(\gamma_i\hat{\alpha}_i)$, needs the following inequality:

$$\begin{aligned} &\left\|\tilde{e}_i^p\,(k_{t+1} + N_i\,(k_{t+1}) - 1 \mid k_{t+1})\right\|_{P_i} \\ &\leq \left\|\bar{e}_i^p\,(k_{t+1} + N_i\,(k_{t+1}) - 1 \mid k_t)\right\|_{P_i} \\ &\quad + \|\tilde{e}_i^p(k_{t+1} + N_i(k_{t+1}) - 1 \mid k_{t+1}) \\ &\quad - \bar{e}_i^p(k_{t+1} + N_i(k_{t+1}) - 1 \mid k_t)\|_{P_i} \\ &\leq \left\|\bar{e}_i^p\,(k_{t+1} + N_i\,(k_{t+1}) - 1 \mid k_t)\right\|_{P_i} + (1 - \gamma_i)\,\hat{\alpha}_i, \end{aligned} \tag{26}$$

where the inequality $\|\tilde{e}_i^p(k_{t+1} + \tau \mid k_{t+1}) - \bar{e}_i^p(k_{t+1} + \tau \mid k_t)\|_{P_i} \leq (1 - \gamma_i)\hat{\alpha}_i$, $\tau \in \mathbb{N}_{[0,N_i(k_0)]}$ can be obtained by combining equation (24) and condition ii).

For $t = 0$, $\bar{e}_i^p(k_1 + N_i(k_1) - 1 \mid k_0) = \bar{e}_i^p(k_0 + N_i(k_0) + \bar{\tau}_d \mid k_0)$. According to $\bar{e}_i^p(k_0 + N_i(k_0) \mid k_0) \in \bar{\Omega}_{P_i}(\gamma_i\hat{\alpha}_i)$, it is easy to proof that $\|\bar{e}_i^p(k_0 + N_i(k_0) + \bar{\tau}_d \mid k_0)\|_{P_i} \leq \gamma_i\hat{\alpha}_i$. In addition, for $t \in \mathbb{N}_{\geq 1}$, $\bar{e}_i^p(k_{t+1} + N_i(k_{t+1}) - 1 \mid k_t) = \bar{e}_i^p(k_t + N_i(k_t) \mid k_t) \in \bar{\Omega}_{P_i}(\gamma_i\hat{\alpha}_i)$.

Hence, it gives $\left\|\tilde{e}_i^p(k_{t+1} + N_i(k_{t+1}) - 1 \mid k_{t+1})\right\|_{P_i} \leq \gamma_i\hat{\alpha}_i + (1 - \gamma_i)\,\hat{\alpha}_i = \hat{\alpha}_i$, which means $\tilde{e}_i^p(k_{t+1} + N_i(k_{t+1}) - 1 \mid k_{t+1}) \in \bar{\Omega}_{P_i}(\hat{\alpha}_i)$. Finally, it can ensure that $\tilde{e}_i^p(k_{t+1} + N_i(k_{t+1}) \mid k_{t+1}) \in \bar{\Omega}_{P_i}(\gamma_i\hat{\alpha}_i)$, for $t \in \mathbb{N}_{\geq 0}$ with condition iii) and following equation

$$\begin{aligned} &\tilde{e}_i^p\,(k_{t+1} + N\,(k_{t+1}) \mid k_{t+1}) \\ &= \tilde{x}_i(k_{t+1} + N_i(k_{t+1}) \mid k_{t+1}) - \bar{x}_0^a(k_{t+1} + N_i(k_{t+1}) \mid k_{t+1}) \\ &= A\tilde{x}_i(k_{t+1} + N_i(k_{t+1}) - 1 \mid k_{t+1}) \\ &\quad + B\bar{u}_0^a(k_{t+1} + N_i(k_{t+1}) - 1 \mid k_{t+1}) \\ &\quad + K(\tilde{x}_i(k_{t+1} + N_i(k_{t+1}) - 1 \mid k_{t+1}) \\ &\quad - \bar{x}_0^a(k_{t+1} + N_i(k_{t+1}) - 1 \mid k_{t+1})) \\ &\quad - A\bar{x}_0^a(k_{t+1} + N_i(k_{t+1}) - 1 \mid k_{t+1}) \end{aligned}$$

$$- B\bar{u}_0^a(k_{t+1}+N_i(k_{t+1})-1 \mid k_{t+1})$$
$$= A_K\tilde{e}_i^p(k_{t+1}+N_i(k_{t+1})-1 \mid k_{t+1}). \tag{27}$$

Therefore, the recursive feasibility of problem **OP** is guaranteed with the feasible control sequence $\tilde{U}_i(k_t)$. ■

In vehicle platooning scenarios, it is unlikely for the leader to frequently change its control plan, and it is even less likely to continuously satisfy the triggering condition (16). Consequently, with the triggering mechanism established in this work, the prediction horizon of the controller is expected to gradually reduce to one step within a finite time period, in accordance with equation (20). Therefore, the following related assumption is made here.

Assumption 5: There exists an integer $\bar{t}_N \in \mathbb{N}_{\geq 1}$, $N_i(k_t) = N_i(k_{t+1}) = 1$, $\forall t \geq \bar{t}_N$.

The following will introduce a shrinking terminal region with parameter $\alpha_i(N_i(k_t))$ for the optimization problem. Then, the closed-loop UUB stability for e_i of the follower system is analyzed as follows.

Theorem 2: For the ith system, $i \in \mathbb{N}_{\geq 1}$, if Theorem 1 and Assumption 5 hold, then the recursive feasibility with a shrinking terminal set and the closed-loop UUB stability are guaranteed under the following conditions:

i) $\mathcal{Z}_{\hat{\delta}_0,\beta}(\tau+\bar{\tau}_d) \subseteq \bar{\Omega}_{P_i}((1-\gamma_i)\tilde{\alpha}_i(\tau))$;

ii) $\check{\alpha}_i \leq \tilde{\alpha}_i(\tau) \leq \alpha_i(\tau) \leq \hat{\alpha}_i$, $\tau \in \mathbb{N}_{[0,N_i(k_0)]}$.

Proof: At the initial time $k_0 = 0$, define $\hat{\alpha}_i = \alpha_i(N_i(k_0)) = \tilde{\alpha}_i(N_i(k_0))$. Then, using the $\bar{\Omega}_{P_i}(\gamma_i\alpha_i(N_i(k_t)))$ as a new terminal set for problem **OP** at time k_t, $\forall t \in \mathbb{N}_{\geq 1}$, in which the $\alpha_i(N_i(k_t))$ is designed as

$$\alpha_i(N_i(k_t)) = \tilde{\alpha}_i(N_i(k_t)) + \sum_{r=1}^{t}\gamma_i^r(\tilde{\alpha}_i(N_i(k_{t-r})) - \tilde{\alpha}_i(N_i(k_{t-r+1}))). \tag{28}$$

Use a similar proof procedure in Theorem 1, and combine the equations (26) and (28) to get

$$\begin{aligned}&\left\|\tilde{e}_i^p(k_{t+1}+N(k_{t+1})-1 \mid k_{t+1})\right\|_{P_i}\\ &\leq \left\|\bar{e}_i^p(k_{t+1}+N_i(k_{t+1})-1 \mid k_t)\right\|_{P_i} + (1-\gamma_i)\tilde{\alpha}_i(N_i(k_{t+1}))\\ &\leq \gamma_i\alpha_i(N_i(k_t)) + (1-\gamma_i)\tilde{\alpha}_i(N_i(k_{t+1}))\\ &= \alpha_i(N_i(k_{t+1})).\end{aligned} \tag{29}$$

Then, one can ensure that $\tilde{e}_i^p(k_{t+1}+N_i(k_{t+1}) \mid k_{t+1}) \in \bar{\Omega}_{P_i}(\gamma_i\alpha_i(N_i(k_{t+1})))$, $\forall t \in \mathbb{N}_{\geq 0}$, with condition iii) in Theorem 1 and equation (27).

At this point, the recursive feasibility for problem **OP** is proved with the new shrinking terminal region, whose size is affected by the shrinking prediction horizon.

Under Assumption 5, for $t \in \mathbb{N}_{\geq \bar{t}_N+1}$, one can obtain following inequality:

$$\begin{aligned}&\alpha_i(N_i(k_t))\\ &= \tilde{\alpha}_i(N_i(k_t)) + \sum_{r=1}^{t}\gamma_i^r(\tilde{\alpha}_i(N_i(k_{t-r})) - \tilde{\alpha}_i(N_i(k_{t-r+1})))\\ &< \tilde{\alpha}_i(1) + \sum_{r=1}^{t-\bar{t}_N}\gamma_i^r(\tilde{\alpha}_i(N_i(k_{t-r})) - \tilde{\alpha}_i(N_i(k_{t-r+1})))\\ &\quad + \gamma_i^{t-\bar{t}_N+1}\sum_{r=t-\bar{t}_N+1}^{t}(\tilde{\alpha}_i(N_i(k_{t-r})) - \tilde{\alpha}_i(N_i(k_{t-r+1})))\\ &= \tilde{\alpha}_i(1) + \gamma_i^{t-\bar{t}_N+1}\left(\tilde{\alpha}_i(N_i(k_0)) - \tilde{\alpha}_i(N_i(k_{\bar{t}_N}))\right)\end{aligned} \tag{30}$$

Since $\gamma_i \in (0, 1)$, it ensures that

$$\lim_{t\to\infty}\alpha_i(N_i(k_t)) < \tilde{\alpha}_i(1) \tag{31}$$

Therefore, there exists an integer $\hat{t}_N \in \mathbb{N}_{\geq \bar{t}_N+1}$ such that

$$\tilde{e}_i^p(k_t) \in \bar{\Omega}_{P_i}(\gamma_i\alpha_i(N_i(k_t))) \subset \bar{\Omega}_{P_i}(\gamma_i\tilde{\alpha}_i(1)),\ t \in \mathbb{N}_{\geq \hat{t}_N} \tag{32}$$

The parameter $\tilde{\alpha}_i$ and its minimum value $\check{\alpha}_i$ can be calculated based on condition i) as

$$\check{\alpha}_i = \tilde{\alpha}_i(1) = \frac{\max_{\rho\in\mathcal{Z}_{\hat{\delta}_0,\beta}(1+\bar{\tau}_d)}\|\rho\|_{P_i}}{(1-\gamma_i)} \tag{33}$$

which demonstrates that the platoon error decreases as the prediction horizon and terminal region shrink. This can ensure closed-loop stability with Algorithm 1 and lead to more precise tracking of the desired trajectories for nominal platoon system. ■

As for the actual system, with the actual control input defined in (10) and the constraints relations between the nominal system and the actual system in (14), one can obtain the invariant set of the control objective in (6) as $\sigma_i = \bar{\Omega}_{P_i}(\gamma_i\tilde{\alpha}_i(1)) \oplus \mathcal{Z}_{\mathcal{E}_i}(1, \bar{\tau}_d)$ for e_i and give $\hat{k} = k_{\hat{t}_N}$.

Remark 3: From the analysis and proofs presented above, it is clear that the range to which the platoon error is ultimately bounded is specifically influenced by the bounded range $\mathcal{W}_i$ of external disturbances, the upper bound $\bar{\tau}_d$ of communication delays, the allowable variation magnitude parameter $\delta_0(\tau)$ of the leader's control plan, and the triggering threshold parameter β. In other words, if we set the final target range for the platoon error to be within a certain limit, it is possible to determine the allowable upper bounds for certain parameters based on other fixed parameters. For instance, given a final target range $\sigma_i = \bar{\Omega}_{P_i}(\gamma_i\check{\rho}_i)$ with $\check{\rho}_i \in (0, \infty)$, the allowable upper bound for communication delays can be determined by solving the following problem:

$$\begin{aligned}&\max_{\bar{\tau}_d\in\mathbb{N}_{\geq 1},\check{\alpha}_i\in(0,\check{\rho}_i)} \bar{\tau}_d\\ &\text{s.t. } \mathcal{Z}_{\mathcal{E}_i}(1,\bar{\tau}_d) \subseteq \bar{\Omega}_{P_i}(\gamma_i(\check{\rho}_i - \check{\alpha}_i))\\ &\qquad \mathcal{Z}_{\hat{\delta}_0,\beta}(1+\bar{\tau}_d) \subseteq \bar{\Omega}_{P_i}((1-\gamma_i)\check{\alpha}_i).\end{aligned} \tag{34}$$

V. Simulation Results

In this section, series of numerical simulations for a platoon consist of one leading vehicle and five following vehicles are designed, to verify the effectiveness of Algorithm 1. The numerical simulations were all run on identical hardware, an i9-13900k 3.0GHz PC with 32GB RAM, using MATLAB 2023a.

TABLE I
ENVIRONMENTAL SETTINGS

Parameters	Values	Parameters	Values
Δt	0.05	$N_i(k_0)$	15
$\bar{\tau}_d$	6	d_0 [m]	20
p_b [m]	5	v_b [m/s]	10
$v_{\min}$ [m/s]	0	$v_{\max}$ [m/s]	30
$a_{\min}$ [m/s^2]	-6	$a_{\max}$ [m/s^2]	6
m_i [kg]	2200	θ_i [rad]	$(-0.04, 0.04)$
$C_{A,i}$	0.3	c_r	0.014
β	0.25	γ_i	0.935
Q_i	diag(10,1)	R_i	1
K	[2.96 2.68]	P_i	[31.9,14.5;14.5,10.8]

A. Comparison I

In the first comparison, the advantages of this method over robust DMPC were assessed. The communication topology of the platoon was leader-following (LF), with each follower receiving leader information subject to the same communication delay $\bar{\tau}_d = 6$.

The parameters of the simulation environment are set in Table I. The initial prediction horizon is set as $N_i(k_0) = 15$, which is long enough to control the initial error $\bar{e}_i^p(k_0)$ into the terminal region $\bar{\Omega}_{P_i}(\gamma_i\bar{\alpha}_i(N_i(k_0)))$. The parameters $v_{\min}$, $v_{\max}$, $a_{\min}$ and $a_{\max}$ are used to describe the system state constraint $\mathcal{X}_i$ and control input constraint $\mathcal{U}_i$ as

$$\mathcal{X}_i = ((-\infty, \infty), [0, 30])^\top, \quad \mathcal{U}_i = [-6, 6].$$

The desired following spacing d_0 is constant, but the velocity of the leader is time-varying, as shown in Figure 3 due to the change of its control plan. The bound associated with the change of leader's control plan is set as $\delta_0(\tau) = 0.15 \times 0.97^\tau$. It is remarkable that the leader's time-varying nominal control inputs $\bar{u}_0(k)$, which are also unknown to the leader itself until they are implemented at time k, are used to simulate a realistic time-varying control plan in an unexpected scenario. Therefore, such change cannot be communicated to the followers in advance through information transmission and be used by them in predictive models to compensate for the information errors due to communication delays.

Figure 3 demonstrates the platoon control performance using the proposed method. It is evident that the platoon error, represented by the deviation between the ith vehicle and the leader vehicle, is maintained within a near-zero range. This indicates that, despite the presence of external disturbances and communication delays, the method ensures the error remains bounded, showcasing its robustness and effectiveness. The control inputs for each vehicle in the platoon are well within the specified constraints, with no abrupt changes or fluctuations observed during the control process. This stability is crucial for maintaining the platoon's integrity and ensuring safety.

In contrast, Figure 4 shows the platoon control performance using the Robust DMPC method without compensation process (5). The graph indicates a significant static error,

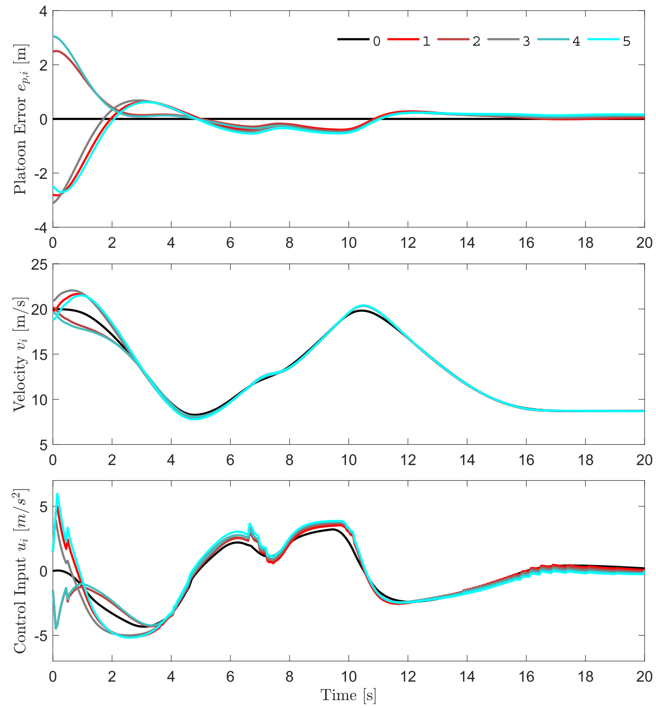

Fig. 3. Platoon control performance using proposed method.

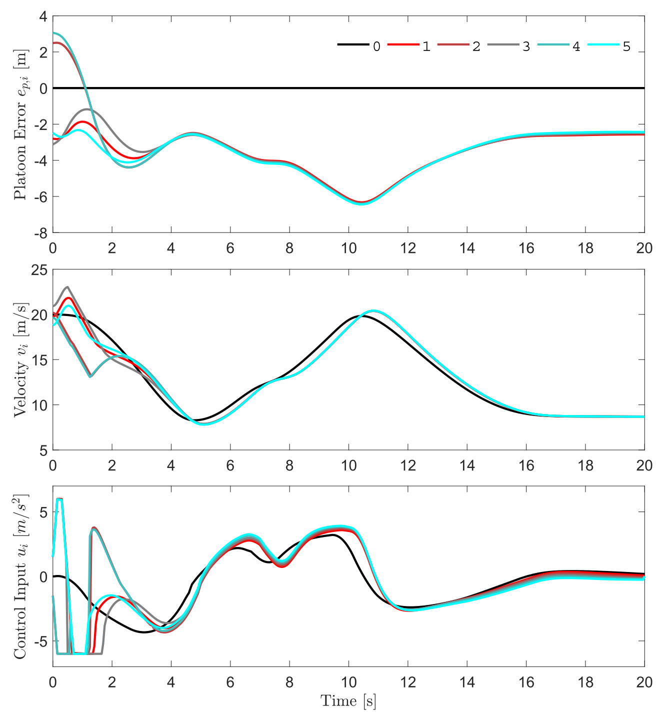

Fig. 4. Platoon control performance using Robust DMPC without compensation (5).

with the platoon error not converging to the vicinity of zero. This underscores the effectiveness and necessity of utilizing timestamp information in MPC under communication delays.

Table II presents a quantitative comparison of the average platoon error (measured by the Mean Absolute Error (MAE) and the Root Mean Square Error (RMSE)), average

TABLE II
PERFORMANCE COMPARISON OF PLATOON CONTROL METHODS

LF topology	Average platoon error [m]		Average spacing error [m]		Average execution time [ms]	Average number of communications
	MAE	RMSE	MAE	RMSE		
Proposed Method	0.354	0.626	0.377	1.036	2.25	262
Robust DMPC	3.576	3.781	0.982	1.587	10.48	325

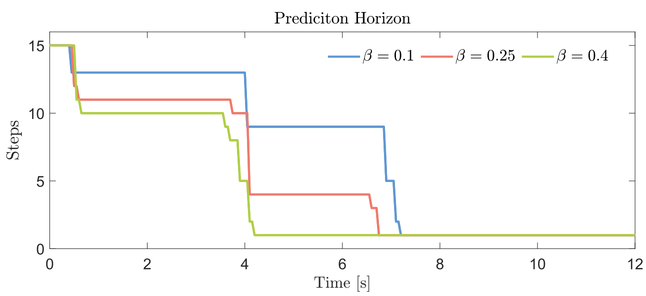

Fig. 5. Prediction horizon shrinking process with different β.

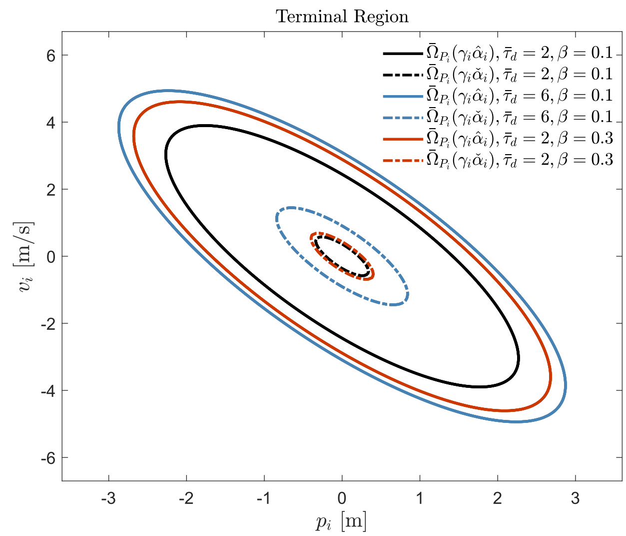

Fig. 6. Terminal sets with different $\bar{\tau}_d$ and β.

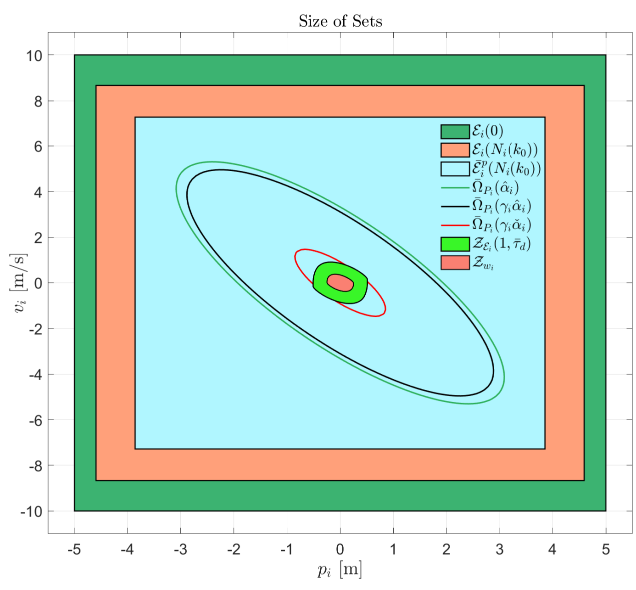

Fig. 7. Size relation among the different sets.

spacing error, average execution time, and number of communications for both methods. Spacing error refers to the discrepancy between the actual interval of a vehicle and its preceding vehicle and the desired interval, denoted as $e_{as,i} = p_i - p_{i-1} + d_0$. The proposed method outperforms the Robust DMPC method in all metrics. Specifically, the average platoon error is substantially lower for the proposed method, with an MAE of 0.354 m and an RMSE of 0.626 m, compared to the Robust DMPC method's MAE of 3.576 m and RMSE of 3.781 m. The proposed method features a shrinking prediction horizon that adapts to changes in the leader's control plan, as depicted in Figure 5, facilitating more efficient communication and optimization. Additionally, the simulation designed a total of 320 control plan changes for the leader, but under the proposed triggering mechanism (16), the frequency of communication due to these plan changes was effectively reduced. This is reflected in the reduced number of communications (262 times) compared to the Robust DMPC method (325 times), which corresponds to a 19.4% decrease. Therefore, the average execution time for the proposed method (2.25 ms) is significantly lower than that of the Robust DMPC method (10.48 ms), highlighting the communication and computational efficiency of the proposed approach.

In Figure 5, one can see the shrinking process of the prediction horizon for different values of β, which will significantly affect the triggering threshold $\mathcal{Z}_\beta$ as demonstrated in equation (12). This result demonstrates that the smaller β is taken, the slower $N_i(k_t)$ decreases, due to the fact that the decrease in the threshold causes the triggering to occur more frequently.

As the prediction horizon shortens, the terminal region can also shrink accordingly, and the corresponding maximum and minimum size of the terminal region can be seen in Figure 6. Combined with the result of Theorem 2, this indicates that the platoon error between the follower and the leader gradually drive to a small neighborhood of zero in finite time. In addition, Figure 6 shows the influence of τ_d and β on the size of the terminal set. In conjunction with Figure 5, it can be observed that an increased value of β reduces the frequency of communication, thereby further decreasing the communication and computational burden. However, it also leads to an enlargement of the terminal domain.

Figure 7 shows the size relation among the different sets mentioned in the controller design and theorem analysis.

TABLE III
ENVIRONMENTAL SETTINGS

Parameters	Values	Parameters	Values
$\bar{\tau}_d$	2	γ_i	0.946
Q_i	$\begin{pmatrix} 10 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	P_i	$\begin{pmatrix} 40.5 & 34.2 & 10.5 \\ 34.2 & 34.7 & 10.9 \\ 10.5 & 10.9 & 4.2 \end{pmatrix}$
ς_i	0.25	K	[2.86 3.49 0.89]

The two largest boxes are hard constraints on actual platoon errors during control, which ensure the basic safety of the platoon by limiting the state errors among vehicles. The subsets within them are for nominal systems, which can be computed by (14) and the constraint relations in Theorems 1 and 2. The set $\mathcal{Z}_{w_i}$ represents the error invariant between the actual system and the nominal system caused by the lumped disturbance. It can be computed based on Equation (11) and the set of lumped disturbances $\mathcal{W}_i$. The latter is estimated from the primary disturbance $f_i(v_i, \theta_i)$ and the related parameter range in Table I, yielding an estimated range $\mathcal{W}_i = [-0.7, 0.7]$.

B. Comparison II

In this comparison, the method was contrasted with the distributed control method incorporating timestamp information as proposed in [29], and the control effectiveness under different information transfer contents within a predecessor-following (PF) communication topology was evaluated. Specifically, this includes the PF topology transmitting the predecessor's state information (PF-P), and the PF topology transmitting the leader's state information (PF-L). Each information transmission within the communication topology involves a delay $\bar{\tau}_d = 2$, at which point the PF-L topology is theoretically equivalent to a LF topology where the delay increases linearly with the distance from the leader ($\bar{\tau}_d = 2 \sim 10$).

Furthermore, to demonstrate the applicability of the method presented in this paper, the model employed in these simulations is the third-order longitudinal dynamic model with the power-train time constant ς_i, as described in work [29]:

$$\dot{x}_i(t) = A_i x_i(t) + B_i u_i(t)$$

with desired acceleration input $u_i(t)$ and matrices

$$A_i = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -\frac{1}{\varsigma_i} \end{bmatrix}, B_i = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\varsigma_i} \end{bmatrix}, x_i(t) = \begin{bmatrix} p_i(t) \\ v_i(t) \\ a_i(t) \end{bmatrix}.$$

The parameter settings that differ from the previous comparison are presented in Table III.

Figure 8 showcases the platoon control performance under PF-P topology using proposed method. It is evident that the method still effectively maintains a stable platoon formation with the new dynamic model, even in the presence of communication delays and disturbance. The controlled platoon exhibits a smooth and consistent response, indicative of robust control under dynamic variations.

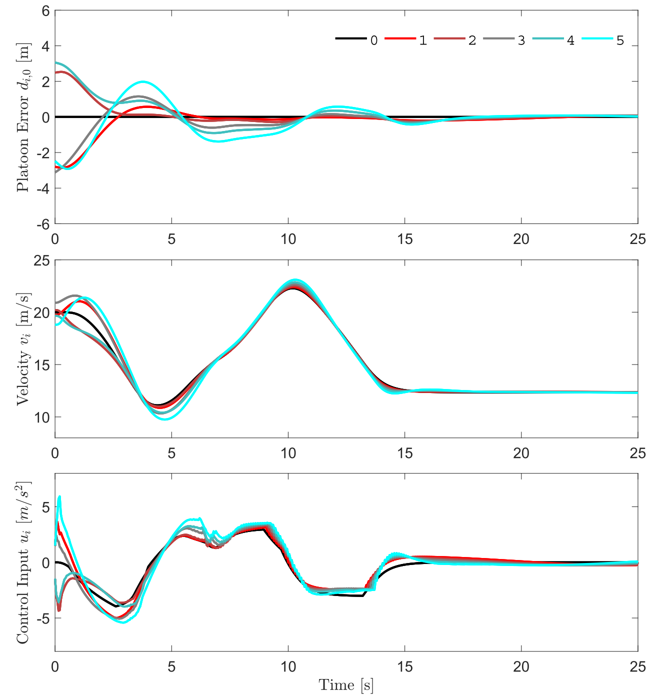

Fig. 8. Platoon control performance using proposed method.

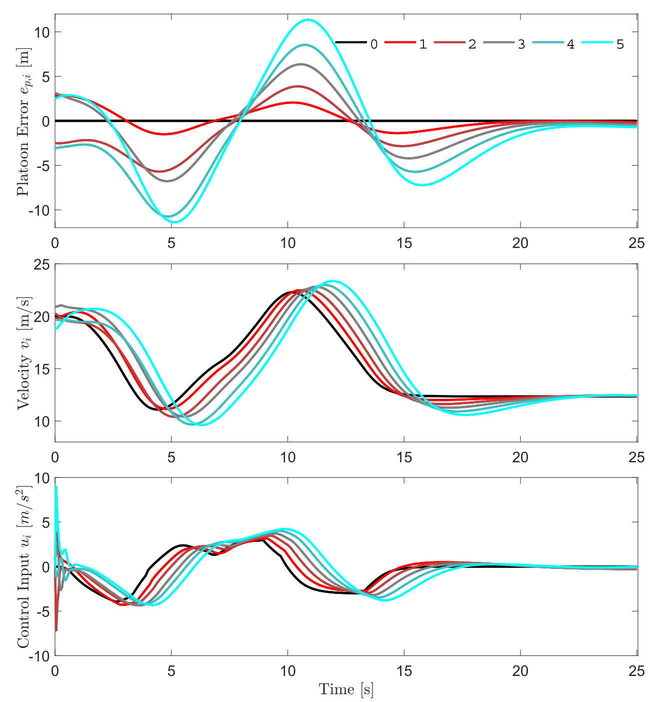

Fig. 9. Platoon control performance using method in [29].

Figure 9, on the other hand, illustrates the performance of the method from [29]. While this method also demonstrates the ability to control the platoon and utilizes timestamp information to ultimately drive the platoon error to near zero, it exhibits a higher tracking error and longer settling

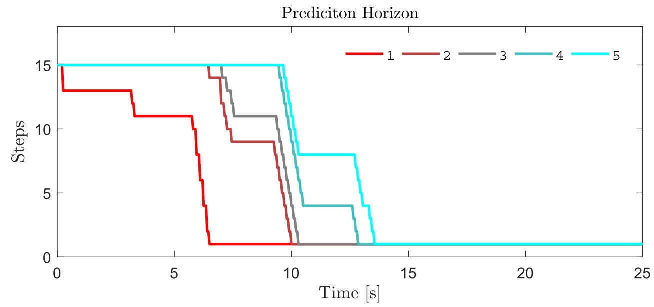

Fig. 10. Prediction horizon shrinking process with PF-P topology.

time. This could be attributed to the differences in control strategy, particularly in how it handles delay compensation. Furthermore, this method does not explicitly address state and control constraints, resulting in violations of constraint boundaries during the control process, whereas the method proposed in this paper performs very well in handling such constraints.

Table IV presents a performance comparison of both methods. Notably, our proposed method exhibits a lower average platoon error and average spacing error, highlighting improved tracking accuracy and platoon stability. Despite having a higher average execution time of 7.33 ms, it is important to note that this value is significantly lower than the 50 ms sampling interval, ensuring that the control updates are well within the real-time constraints. Moreover, our method demonstrates a substantial reduction in communication frequency, a critical advantage in reducing network load and enhancing the overall system's resilience. The lower communication frequency can also contribute to lower energy consumption and increased scalability of the platoon.

It is worth mentioning that the average computation time in this simulation is significantly longer than that in the LF topology simulations. This is because the triggering of information transmission now depends on the plan changes of the preceding vehicle rather than the leader, resulting in a longer process for the corresponding shrinking of the prediction horizon, as detailed in Figure 10.

Switching the platoon's topology to PF-L, the simulation results for both methods are presented in Figures 11 and 12. At this point, the platoon error control performance of the method from [29] has improved significantly compared to the results under the PF-P topology, but the method proposed in this paper still demonstrates noticeably better control performance and constraint management.

The specific quantitative comparison results can be seen in Table V. The error control effect of proposed method remained essentially unchanged, while the average execution time and the number of communications both decreased. The enhancement in computational and communication efficiency is attributed to the change in the content of information transmission to leader information. Consequently, the condition for triggering information transmission is solely influenced by the extent of changes in the leader's control plan and is independent of the preceding vehicle's control process. This can be substantiated by observing the roughly identical shrinking

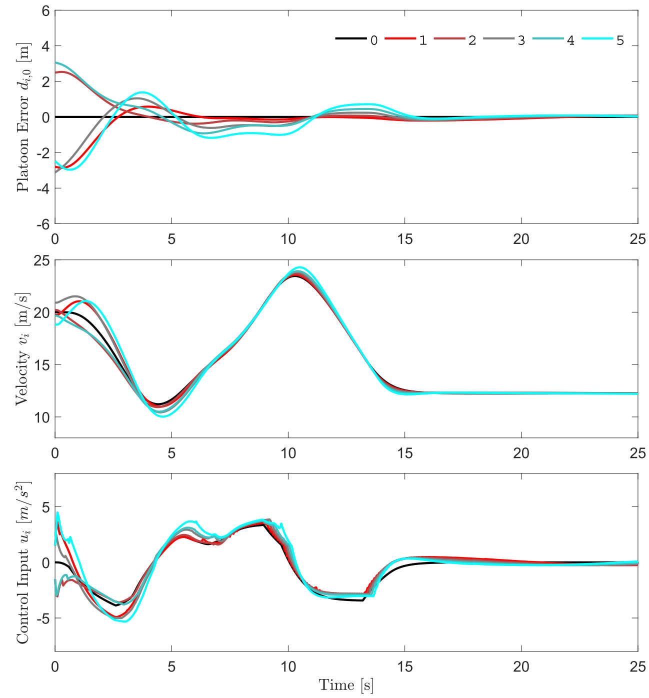

Fig. 11. Platoon control performance using proposed method.

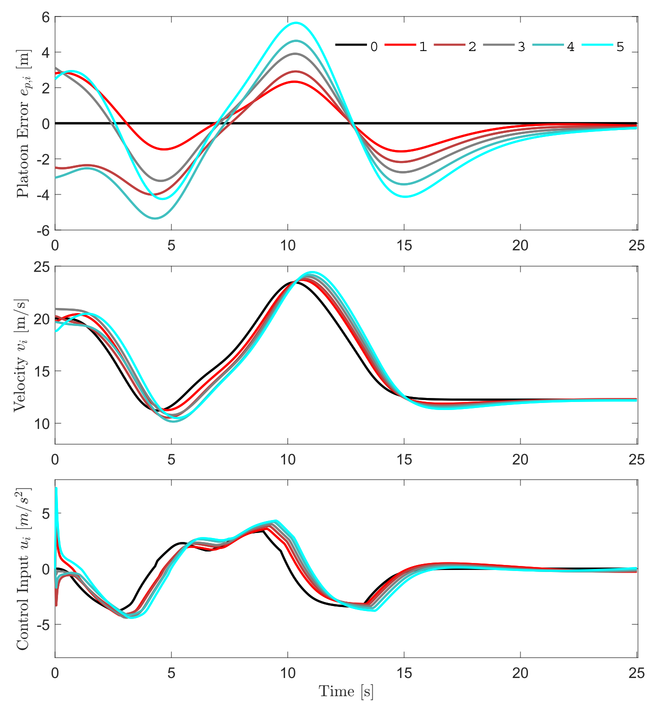

Fig. 12. Platoon control performance using method in [29].

processes of the predicted horizons in Figure 13. In addition, the differences between various followers in Figure 13 are due to the different total communication delays caused by the different number of transmission times each vehicle receives the leader information.

The simulation results and performance comparison confirm the superiority of the proposed platoon control method.

TABLE IV
PERFORMANCE COMPARISON OF PLATOON CONTROL METHODS

PF-P topology	Average platoon error [m]		Average spacing error [m]		Average execution time [ms]	Average number of communications
	MAE	RMSE	MAE	RMSE		
Proposed Method	0.411	0.721	0.420	1.092	7.33	314
Method in [29]	2.793	3.546	1.412	1.926	0.02	500

TABLE V
PERFORMANCE COMPARISON OF PLATOON CONTROL METHODS

PF-L topology	Average platoon error [m]		Average spacing error [m]		Average execution time [ms]	Average number of communications
	MAE	RMSE	MAE	RMSE		
Proposed Method	0.407	0.722	0.435	1.120	4.24	276
Method in [29]	1.727	2.117	1.014	1.636	0.02	500

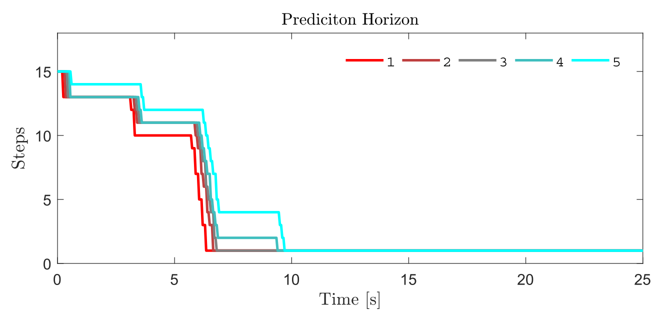

Fig. 13. Prediction horizon shrinking process with PF-L topology.

The method's ability to drive the errors to near-zero, optimize communication frequency, and reduce computation time demonstrates its potential for real-world applications where precision and efficiency are paramount.

C. Simulation III

In this simulation, the robustness and adaptability of the proposed method is further evaluated under different control tasks and initial conditions. The simulation setup is consistent with Comparison II, except for the topology, which is changed to a LF structure, and the initial prediction horizon is increased to 30 steps. The simulation introduces a sudden change in the desired platoon spacing at the 15-second mark, reducing the spacing from 20 m to 19 m while the leader simultaneously slows down the platoon speed. This scenario is designed to mimic real-world situations where higher-level intelligent transportation systems or infrastructure may impose new control objectives on the platoon.

The simulation results are presented in Fig. 14 and Fig. 15. Fig. 14 showcases the platoon control performance under the proposed method. It is evident that the method effectively maintains a stable platoon formation even when faced with sudden changes in the desired spacing and leader speed. The controlled platoon exhibits a smooth and consistent response, demonstrating robust control under dynamic variations. At the 15-second mark, when the desired spacing is reduced, the system quickly adapts to the new control target, ensuring that the platoon error converges to near-zero values.

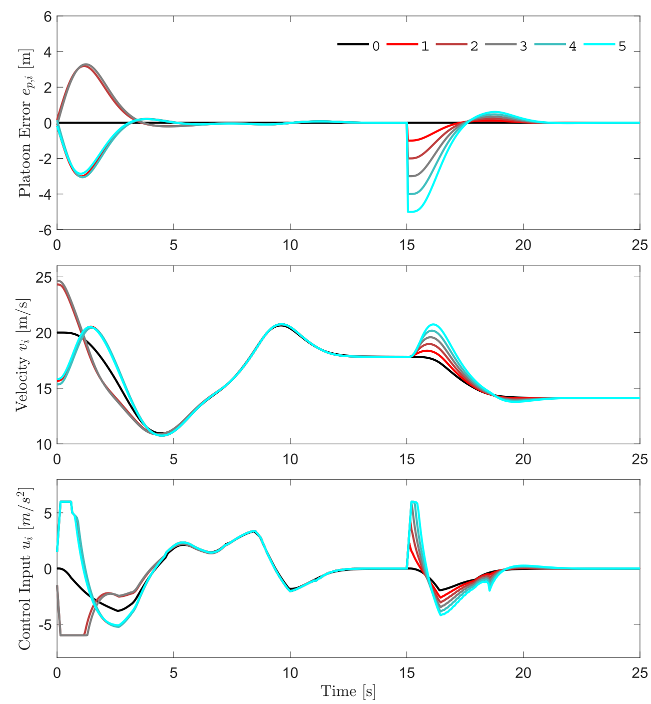

Fig. 14. Platoon control performance with proposed method under different control tasks.

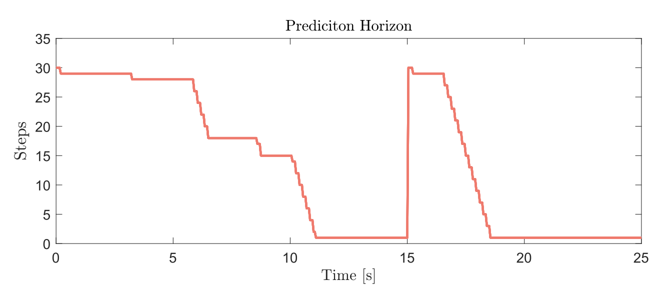

Fig. 15. Prediction horizon shrinking process under different control tasks.

Fig. 15 illustrates the prediction horizon shrinking process under different control tasks. Initially, the system starts with a large prediction horizon of 30 steps to handle the large initial velocity errors. As the state approaches the terminal region, the prediction horizon rapidly decreases to its minimum value. At the 15-second mark, when faced with the new control target, the state error suddenly increases, deviating from the new terminal target. The MPC then reverts to the maximum prediction horizon to ensure feasibility and quickly reduces the prediction horizon to its minimum as the new target is achieved. This dynamic adjustment of the prediction horizon ensures both control effectiveness and computational efficiency. Moreover, the prediction horizon shrinks rapidly as the system converges, indicating that communication is not frequent but still achieves fast and stable control performance. This demonstrates the efficiency of our approach in reducing both communication and computational burdens.

Under the same hardware conditions, we statistically analyzed the simulation scenarios with prediction horizons greater than 25 steps and less than 5 steps. The average computation times for MPC were 48.13 ms and 2.96 ms, respectively. The former computation time is close to the MPC sampling interval of 0.05 s, while the latter reduces the computation time by 93.85%, providing ample computational margin for unexpected situations and control smoothness during the system control process. These simulation results demonstrate the adaptability of the proposed method, ensuring both control effectiveness and significant improvement in computational efficiency under different control tasks.

The proposed method demonstrates excellent robustness to initial conditions and adaptability to new control tasks. The ability to handle large initial errors and sudden changes in control objectives highlights its potential for real-world applications. Furthermore, the shrinking prediction horizon strategy significantly reduces the computational burden, making the method highly efficient for real-time control systems. These results confirm the superiority of the proposed method in terms of control accuracy, computational efficiency, and communication frequency, making it a promising solution for vehicle platoon control under communication delays and disturbances.

VI. Conclusion

In this work, an event-triggered robust DMPC strategy with shrinking terminal ingredients is presented for vehicle platooning under disturbance and communication delays. The number of information transmission times between leader and follower is reduced by the proposed communication triggering mechanism, and the computational burden is also reduced thanks to the prediction horizon updating strategy. The proposed strategy have ensured the recursive feasibility of the local optimization problem and the UUB stability of the platoon system when considering the leader's control plan changes during the communication delay. The simulation results validate the effectiveness and robustness of the proposed method.

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